Maharashtra Education Society's

Institute of Management and Career Courses (IMCC), Pune

AUTONOMOUS

A.Y. 2025-26

Date:-21/4/2026

MCA-I Sem-II IT-21-Java Programming

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Assignment 1: Create a Login System

- **Description**: Build a login screen using Java Swing, where the user can enter their username and password. Upon clicking the "Login" button:
    - The application should validate the credentials by checking the user information stored in a database (e.g., MySQL or SQLite).
    - If the credentials are correct, show a success message and navigate to a new screen (or show a new panel).
    - If the credentials are incorrect, display an error message.

```
import javax.swing.JFrame;

import javax.swing.JLabel;

import javax.swing.JOptionPane;
```

```
import java.awt.EventQueue;
import java.awt.Font;
import javax.swing.JTextField;
import javax.swing.JButton;
import java.awt.event.ActionListener;
import java.sql.Connection;
import java.sql.DriverManager;
import java.sql.PreparedStatement;
import java.sql.ResultSet;
import java.awt.event.ActionEvent;

public class Loginform extends JFrame {

	private static final long serialVersionUID = 1L;
	private JTextField textField;
	private JTextField textField_1;

	/**
	 * Launch the application.
	 */
	public static void main(String[] args) {
		EventQueue.invokeLater(new Runnable() {
			public void run() {
```

```
                    try {
                        Loginform frame = new Loginform();
                        frame.setVisible(true);
                    } catch (Exception e) {
                        e.printStackTrace();
                    }
                }
            });
}

/**
 * Create the frame.
 */
public Loginform() {
        setDefaultCloseOperation(JFrame.EXIT_ON_CLOSE);
        setBounds(100, 100, 450, 300);
        getContentPane().setLayout(null);

        JLabel lblNewLabel = new JLabel("Login:");
        lblNewLabel.setFont(new Font("Tahoma", Font.BOLD, 13));
        lblNewLabel.setBounds(94, 32, 84, 12);
        getContentPane().add(lblNewLabel);
```

```
JLabel lblNewLabel_1 = new JLabel("Name:");
lblNewLabel_1.setBounds(94, 66, 44, 12);
getContentPane().add(lblNewLabel_1);

JLabel lblNewLabel_2 = new JLabel("password:");
lblNewLabel_2.setBounds(94, 88, 65, 12);
getContentPane().add(lblNewLabel_2);

textField = new JTextField();
textField.setBounds(154, 63, 96, 18);
getContentPane().add(textField);
textField.setColumns(10);

textField_1 = new JTextField();
textField_1.setBounds(154, 85, 96, 18);
getContentPane().add(textField_1);
textField_1.setColumns(10);

JButton btnNewButton = new JButton("Login");
btnNewButton.addActionListener(new ActionListener() {
        public void actionPerformed(ActionEvent e) {
                try {
                        Class.forName("com.mysql.cj.jdbc.Driver");
```

```
String jdbcUrl="jdbc:mysql://localhost:3306/Login";
String dbname="root";
String dbpass="Root01";

Connection con=DriverManager.getConnection(jdbcUrl,dbname,dbpass);

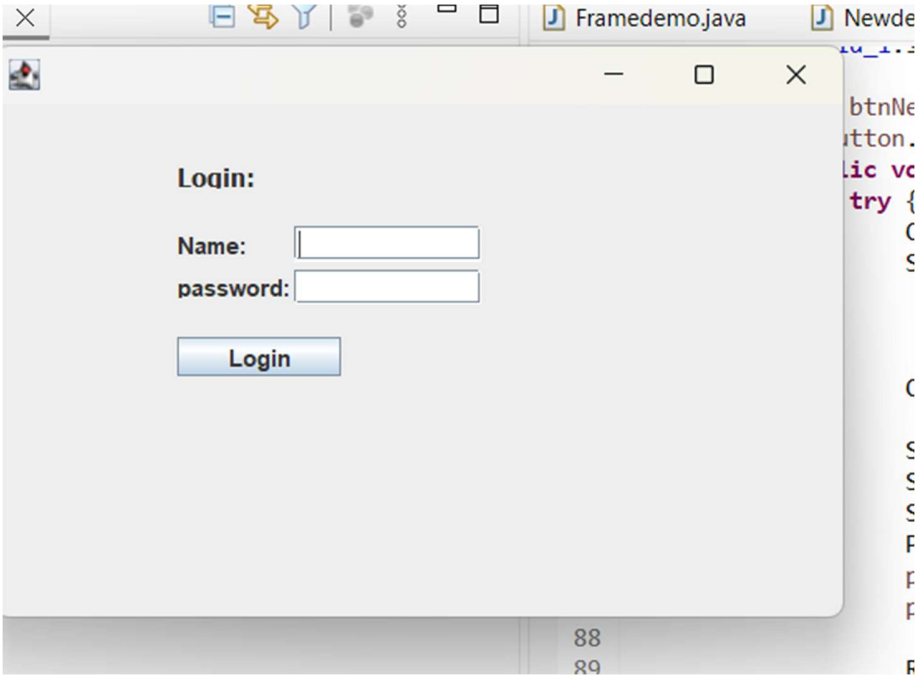
String username=textField.getText();
String password=textField_1.getText();
String query = "SELECT * FROM users WHERE username=? AND password=?";
PreparedStatement pst = con.prepareStatement(query);
pst.setString(1, username);
pst.setString(2, password);

ResultSet rs = pst.executeQuery();
if (rs.next()) {
   JOptionPane.showMessageDialog( null, "Login Successful!");

} else {
   JOptionPane.showMessageDialog(null, "Invalid Username or Password!");
}
```

```
                    }catch(Exception e1) {
                        e1.printStackTrace();
                    }
                }
            });
            btnNewButton.setBounds(94, 120, 84, 20);
            getContentPane().add(btnNewButton);

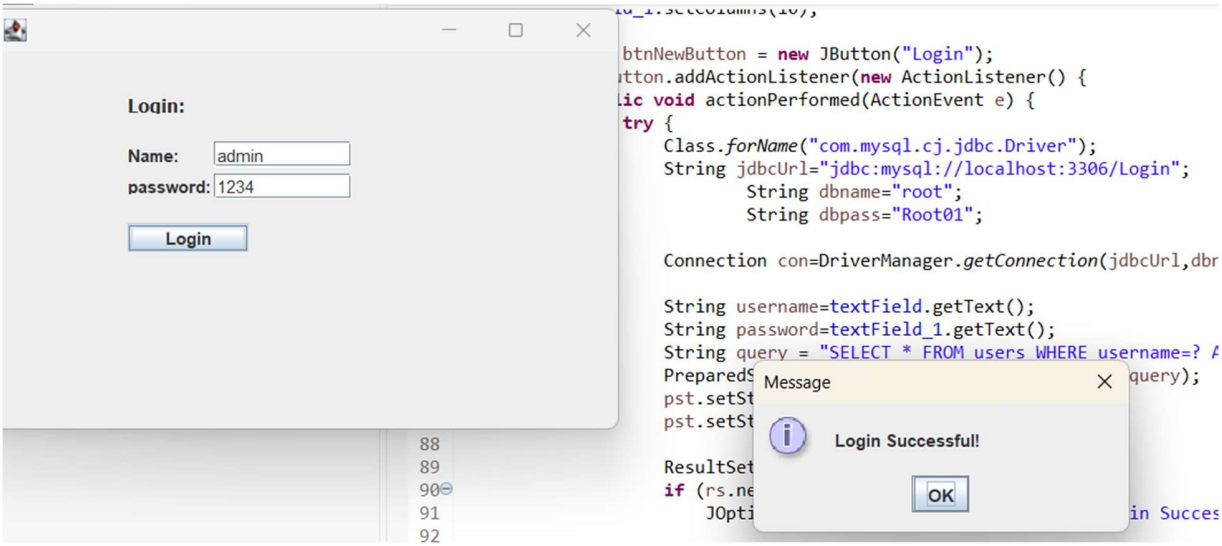
        }
}
```



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Assignment 2: CRUD Operations for a Student Management System

- **Description**: Develop a simple student management system with the following features:
  - Add new student details (name, roll number, course).
  - View the list of students in a JTable.
  - Update student details.
  - Delete student records.

```
import java.awt.EventQueue;


import javax.swing.JFrame;
import javax.swing.JOptionPane;
import javax.swing.JPanel;
import javax.swing.border.EmptyBorder;
import javax.swing.JButton;
import javax.swing.JTable;
import javax.swing.JScrollPane;
import java.awt.event.ActionListener;
import java.sql.Connection;
import java.sql.DriverManager;
import java.sql.PreparedStatement;
import java.sql.ResultSet;
import java.awt.event.ActionEvent;
import javax.swing.table.DefaultTableModel;


import net.proteanit.sql.DbUtils;


import javax.swing.JLabel;
import javax.swing.JTextField;


public class Crud_record extends JFrame {
```

```
private static final long serialVersionUID = 1L;
private JPanel contentPane;
private JTable table;
private JTextField textField;
private JTextField textField_1;
private JTextField textField_2;


/**
 * Launch the application.
 */
public static void main(String[] args) {
	EventQueue.invokeLater(new Runnable() {
		public void run() {
			try {
				Crud_record frame = new Crud_record();
				frame.setVisible(true);
			} catch (Exception e) {
				e.printStackTrace();
			}
		}
	});
}
```

```
/**
 * Create the frame.
 */
public Crud_record() {
        setDefaultCloseOperation(JFrame.EXIT_ON_CLOSE);
        setBounds(100, 100, 450, 300);
        contentPane = new JPanel();
        contentPane.setBorder(new EmptyBorder(5, 5, 5, 5));
        setContentPane(contentPane);
        contentPane.setLayout(null);

        JButton btnNewButton = new JButton("add");
        btnNewButton.addActionListener(new ActionListener() {
                public void actionPerformed(ActionEvent e) {
                        try {
            Connection con = DriverManager.getConnection(
                "jdbc:mysql://localhost:3306/student_db",
                "root",
                "Root01");

            String sql = "INSERT INTO student(name, roll, course) VALUES (?, ?, ?)";
            PreparedStatement pst = con.prepareStatement(sql);
```

```
        pst.setString(1, textField.getText());
        pst.setString(2, textField_1.getText());
        pst.setString(3, textField_2.getText());

        pst.executeUpdate();

        JOptionPane.showMessageDialog(null, "Student Added");


    } catch (Exception e1) {
        e1.printStackTrace();
    }
}

 });
 btnNewButton.setBounds(277, 144, 84, 20);
 contentPane.add(btnNewButton);

 table = new JTable();
 table.setModel(new DefaultTableModel(
            new Object[][] {
            },
```

```
		new String[] {
			"Id:", "Name:", "Roll:", "Course:"
		}
	));
	table.setBounds(65, 174, 305, 89);

	JScrollPane scrollPane = new JScrollPane(table);
	scrollPane.setBounds(65, 191, 305, 62);
	contentPane.add(scrollPane);
	scrollPane.setColumnHeaderView(table.getTableHeader());

	JLabel lblNewLabel = new JLabel("Name:");
	lblNewLabel.setBounds(78, 31, 44, 12);
	contentPane.add(lblNewLabel);

	JLabel lblNewLabel_1 = new JLabel("Roll:");
	lblNewLabel_1.setBounds(78, 53, 44, 12);
	contentPane.add(lblNewLabel_1);

	JLabel lblNewLabel_2 = new JLabel("Course:");
	lblNewLabel_2.setBounds(78, 86, 44, 12);
	contentPane.add(lblNewLabel_2);
```

```
textField = new JTextField();
textField.setBounds(123, 28, 96, 18);
contentPane.add(textField);
textField.setColumns(10);

textField_1 = new JTextField();
textField_1.setBounds(123, 56, 96, 18);
contentPane.add(textField_1);
textField_1.setColumns(10);

textField_2 = new JTextField();
textField_2.setBounds(123, 83, 96, 18);
contentPane.add(textField_2);
textField_2.setColumns(10);

JButton btnNewButton_1 = new JButton("display");
btnNewButton_1.addActionListener(new ActionListener() {
        public void actionPerformed(ActionEvent e) {
                try {
Connection con = DriverManager.getConnection(
 "jdbc:mysql://localhost:3306/student_db",
 "root",
 "Root01");
```

```
            String sql = "select * from student";
            PreparedStatement pst = con.prepareStatement(sql);
            ResultSet rs=pst.executeQuery();
            table.setModel(DbUtils.resultSetToTableModel(rs));

        } catch (Exception e1) {
            e1.printStackTrace();
        }
    }

            });
            btnNewButton_1.setBounds(183, 144, 84, 20);
            contentPane.add(btnNewButton_1);

            JButton btnNewButton_2 = new JButton("update");
            btnNewButton_2.addActionListener(new ActionListener() {
                    public void actionPerformed(ActionEvent e) {

                            int row = table.getSelectedRow();

                            if (row == -1) {
```

```
                JOptionPane.showMessageDialog(null, "Select a row
first!");
                return;
            }

            try {
                DefaultTableModel model = (DefaultTableModel)
table.getModel();
                int id = Integer.parseInt(model.getValueAt(row,
0).toString());

                Connection con = DriverManager.getConnection(
                    "jdbc:mysql://localhost:3306/student_db",
                    "root", "Root01");

                String sql = "UPDATE student SET name=?, roll=?, course=?
WHERE id=?";
                PreparedStatement pst = con.prepareStatement(sql);

                pst.setString(1, textField.getText());
                pst.setInt(2, Integer.parseInt(textField_1.getText()));
                pst.setString(3, textField_2.getText());
                pst.setInt(4, id);
```

```
                    pst.executeUpdate();

                    JOptionPane.showMessageDialog(null, "Updated
Successfully");

                } catch (Exception e1) {
                    e1.printStackTrace();
                }
            }
        });
    btnNewButton_2.setBounds(89, 144, 84, 20);
    contentPane.add(btnNewButton_2);

    JButton btnNewButton_3 = new JButton("delete");
    btnNewButton_3.addActionListener(new ActionListener() {
        public void actionPerformed(ActionEvent e) {
            int row = table.getSelectedRow();

      if (row == -1) {
        JOptionPane.showMessageDialog(null, "Select a row first!");
        return;
      }
```

```
        try {
            DefaultTableModel model = (DefaultTableModel) table.getModel();
            int id = Integer.parseInt(model.getValueAt(row, 0).toString());

            Connection con = DriverManager.getConnection(
                    "jdbc:mysql://localhost:3306/student_db",
                    "root", "Root01");

            String sql = "delete from student where id=?";
            PreparedStatement pst = con.prepareStatement(sql);

            pst.setInt(1, id);
            pst.executeUpdate();

            JOptionPane.showMessageDialog(null, "deleted Successfully");

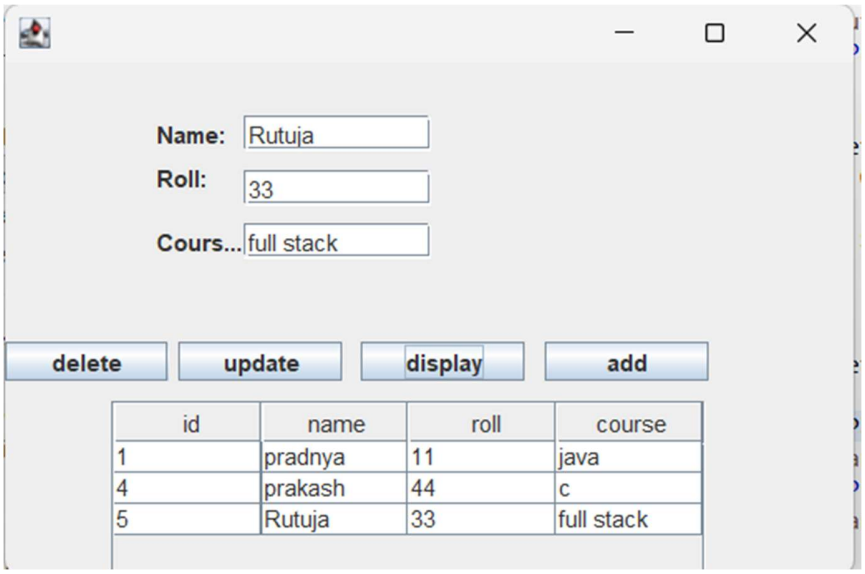
        } catch (Exception e1) {
            e1.printStackTrace();
        }

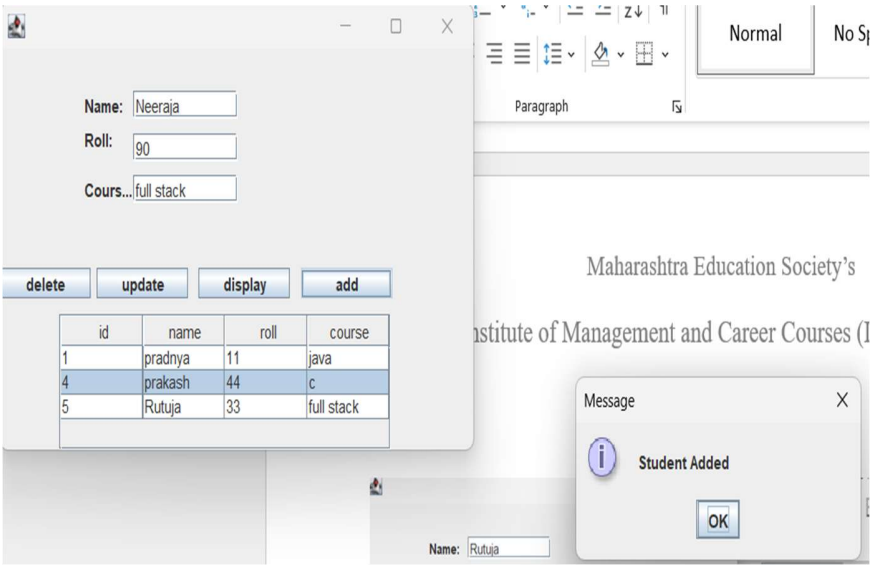
         }
});
btnNewButton_3.setBounds(0, 144, 84, 20);
```

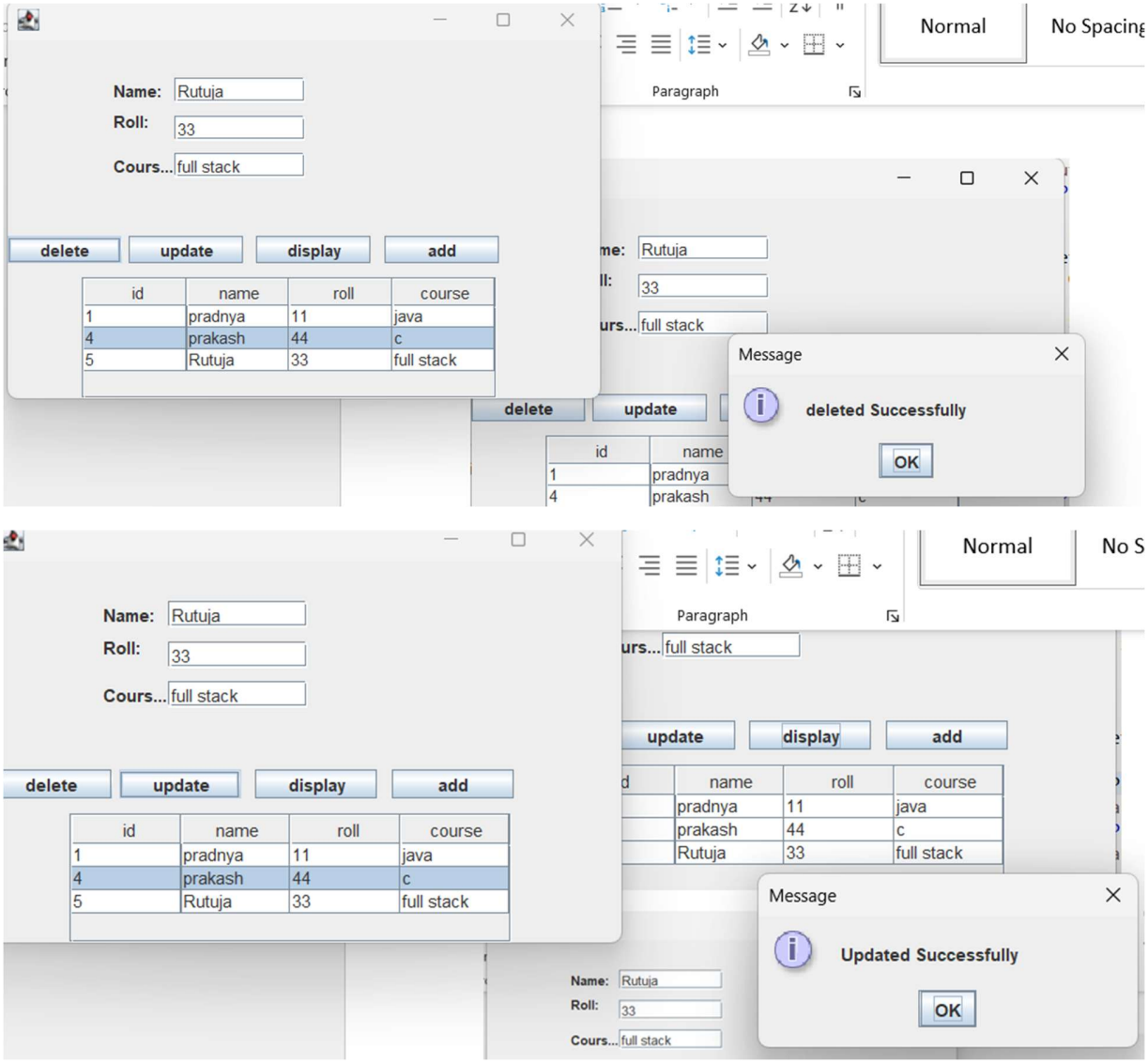
```
        contentPane.add(btnNewButton_3);


    }

}
```







Assignment 3: Library Book Inventory System

- **Description**: Create a library book inventory system that allows the following functionalities:
    - Add a new book with details like title, author, ISBN, and quantity.
    - View available books in the library using a JTable.

- Update the book information (e.g., change the quantity of books).
- Delete a book from the inventory.
- Search for books by title or author.

```
import java.awt.EventQueue;


import javax.swing.JFrame;
import javax.swing.JPanel;
import javax.swing.border.EmptyBorder;
import javax.swing.JLabel;
import javax.swing.JOptionPane;

import java.awt.Font;
import javax.swing.JTextField;
import javax.swing.JButton;
import javax.swing.JScrollPane;
import javax.swing.JTable;
import javax.swing.table.DefaultTableModel;

import net.proteanit.sql.DbUtils;

import java.awt.event.ActionListener;
import java.sql.Connection;
```

```
import java.sql.DriverManager;
import java.sql.PreparedStatement;
import java.sql.ResultSet;
import java.awt.event.ActionEvent;


public class LibrarayFrame extends JFrame {

	private static final long serialVersionUID = 1L;
	private JPanel contentPane;
	private JTextField t1;
	private JTextField t2;
	private JTextField t3;
	private JTextField t4;
	private JTextField searchtxt;
	private JTable table;

	/**
	 * Launch the application.
	 */
	public static void main(String[] args) {
		EventQueue.invokeLater(new Runnable() {
			public void run() {
```

```
                    try {
                        LibrarayFrame frame = new LibrarayFrame();
                        frame.setVisible(true);
                    } catch (Exception e) {
                        e.printStackTrace();
                    }
                }
            });
}

/**
 * Create the frame.
 */
public LibrarayFrame() {
        setDefaultCloseOperation(JFrame.EXIT_ON_CLOSE);
        setBounds(100, 100, 450, 300);
        contentPane = new JPanel();
        contentPane.setBorder(new EmptyBorder(5, 5, 5, 5));
        setContentPane(contentPane);
        contentPane.setLayout(null);

        JLabel lblNewLabel = new JLabel("Library Book Inventory system");
        lblNewLabel.setFont(new Font("Tahoma", Font.BOLD, 13));
```

```
lblNewLabel.setBounds(92, 30, 238, 12);
contentPane.add(lblNewLabel);

JLabel lblNewLabel_1 = new JLabel("title:");
lblNewLabel_1.setBounds(40, 62, 44, 12);
contentPane.add(lblNewLabel_1);

JLabel lblNewLabel_2 = new JLabel("author:");
lblNewLabel_2.setBounds(40, 96, 44, 12);
contentPane.add(lblNewLabel_2);

JLabel lblNewLabel_3 = new JLabel("ISBN:");
lblNewLabel_3.setBounds(40, 128, 44, 12);
contentPane.add(lblNewLabel_3);

JLabel lblNewLabel_4 = new JLabel("quantity:");
lblNewLabel_4.setBounds(40, 158, 44, 12);
contentPane.add(lblNewLabel_4);

t1 = new JTextField();
t1.setBounds(92, 59, 96, 18);
contentPane.add(t1);
t1.setColumns(10);
```

```
t2 = new JTextField();
t2.setBounds(92, 93, 96, 18);
contentPane.add(t2);
t2.setColumns(10);

t3 = new JTextField();
t3.setBounds(92, 125, 96, 18);
contentPane.add(t3);
t3.setColumns(10);

t4 = new JTextField();
t4.setBounds(92, 155, 96, 18);
contentPane.add(t4);
t4.setColumns(10);



JButton btnNewButton_1 = new JButton("update");
btnNewButton_1.addActionListener(new ActionListener() {
   public void actionPerformed(ActionEvent e) {
      try {
```

```
int row = table.getSelectedRow();


if (row == -1) {
   JOptionPane.showMessageDialog(null, "Please select a row first!");
   return;
}


DefaultTableModel model = (DefaultTableModel) table.getModel();


int id = Integer.parseInt(model.getValueAt(row, 0).toString());


Connection con = DriverManager.getConnection(
      "jdbc:mysql://localhost:3306/lib_db",
      "root",
      "Root01");


String sql = "UPDATE lib SET title=?, author=?, isbn=?, quantity=? WHERE
id=?";
```

```
            PreparedStatement pst = con.prepareStatement(sql);


            pst.setString(1, t1.getText());
            pst.setString(2, t2.getText());
            pst.setString(3, t3.getText());
            pst.setInt(4, Integer.parseInt(t4.getText()));
            pst.setInt(5, id);


            pst.executeUpdate();


            JOptionPane.showMessageDialog(null, "Book Updated Successfully!");


        } catch (Exception e1) {
            e1.printStackTrace();
        }
    }
});
btnNewButton_1.setBounds(24, 209, 84, 20);
contentPane.add(btnNewButton_1);

JButton btnNewButton_2 = new JButton("delete");
btnNewButton_2.addActionListener(new ActionListener() {
            public void actionPerformed(ActionEvent e) {
```

```
            try {
int row = table.getSelectedRow();



if (row == -1) {
   JOptionPane.showMessageDialog(null, "Please select a row first!");
   return;
}



DefaultTableModel model = (DefaultTableModel) table.getModel();



int id = Integer.parseInt(model.getValueAt(row, 0).toString());



Connection con = DriverManager.getConnection(
      "jdbc:mysql://localhost:3306/lib_db",
      "root",
      "Root01");



String sql = "delete from lib where id=?";
```

```
            PreparedStatement pst = con.prepareStatement(sql);

            pst.setInt(1, id);

            pst.executeUpdate();

            JOptionPane.showMessageDialog(null, "Book deleted Successfully!");

        } catch (Exception e1) {
            e1.printStackTrace();
        }
    }
});



btnNewButton_2.setBounds(118, 183, 84, 20);
contentPane.add(btnNewButton_2);

JButton btnNewButton_3 = new JButton("search");
btnNewButton_3.addActionListener(new ActionListener() {
        public void actionPerformed(ActionEvent e) {
```

```
        try {
            Connection con = DriverManager.getConnection(
                    "jdbc:mysql://localhost:3306/lib_db",
                    "root", "Root01");

            String keyword = searchtxt.getText();

            String sql = "SELECT * FROM lib WHERE title LIKE ? OR author LIKE ?";
            PreparedStatement pst = con.prepareStatement(sql);

            pst.setString(1, "%" + keyword + "%");
            pst.setString(2, "%" + keyword + "%");

            ResultSet rs = pst.executeQuery();
            table.setModel(net.proteanit.sql.DbUtils.resultSetToTableModel(rs));

        } catch (Exception ex) {
            ex.printStackTrace();
        }

         }
});
btnNewButton_3.setBounds(200, 238, 84, 20);
```

```
        contentPane.add(btnNewButton_3);

        searchtxt = new JTextField();
        searchtxt.setBounds(92, 239, 96, 18);
        contentPane.add(searchtxt);
        searchtxt.setColumns(10);
//      btnNewButton_4.setBounds(118, 209, 84, 20);
//      contentPane.add(btnNewButton_4);



        table = new JTable();
        table.setModel(new DefaultTableModel(
                new Object[][] {
                },
                new String[] {
                        "Id:", "title:", "author:", "ISBN:", "quantity:"
                }
        ));
        table.setBounds(243, 78, 183, 134);


        JScrollPane scrollPane = new JScrollPane(table);
```

```
scrollPane.setBounds(243, 78, 183, 134);
contentPane.add(scrollPane);
scrollPane.setColumnHeaderView(table.getTableHeader());

JLabel lblNewLabel_5 = new JLabel("searchByTitle:");
lblNewLabel_5.setBounds(24, 241, 68, 12);
contentPane.add(lblNewLabel_5);

JButton btnNewButton_5 = new JButton("display");
btnNewButton_5.setBounds(118, 209, 84, 20);
contentPane.add(btnNewButton_5);



JButton btnNewButton = new JButton("add");
btnNewButton.addActionListener(new ActionListener() {
        public void actionPerformed(ActionEvent e) {
                try {
                        Connection con=DriverManager.getConnection(
                                        "jdbc:mysql://localhost:3306/lib_db",
                                        "root",
                                        "Root01");
```

```
                                        String query="insert into
lib(title,author,isbn,quantity)values(?,?,?,?)";
                                        PreparedStatement pst=con.prepareStatement(query);
                                        pst.setString(1,t1.getText());
                                        pst.setString(2,t2.getText());
                                        pst.setString(3,t3.getText());
                                        pst.setInt(4,Integer.parseInt(t4.getText()));
                                        pst.executeUpdate();
                                        JOptionPane.showConfirmDialog(null, "book added!");


                }catch(Exception e1) {
                        e1.printStackTrace();
                }

                }
        });
        btnNewButton.setBounds(24, 183, 84, 20);
        contentPane.add(btnNewButton);

//      JButton btnNewButton_5 = new JButton("display");
        btnNewButton_5.addActionListener(new ActionListener() {
                public void actionPerformed(ActionEvent e) {
```

```
                try {
                Connection con=DriverManager.getConnection(
                                "jdbc:mysql://localhost:3306/lib_db",
                                "root",
                                "Root01");
                String sql="select * from lib";
                PreparedStatement pst=con.prepareStatement(sql);
                ResultSet rs=pst.executeQuery();
                table.setModel(DbUtils.resultSetToTableModel(rs));

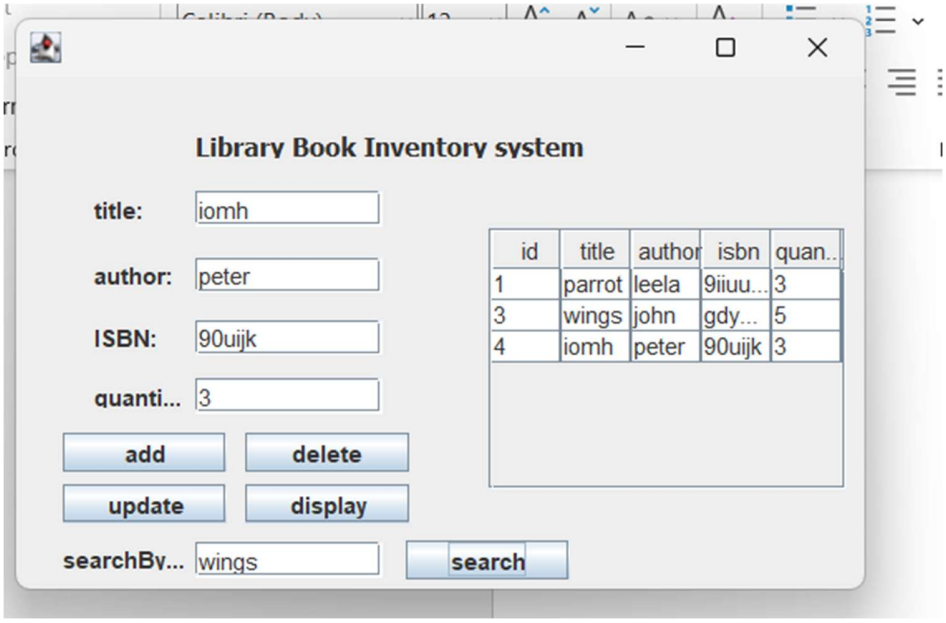
                }catch(Exception e1) {
                        e1.printStackTrace();

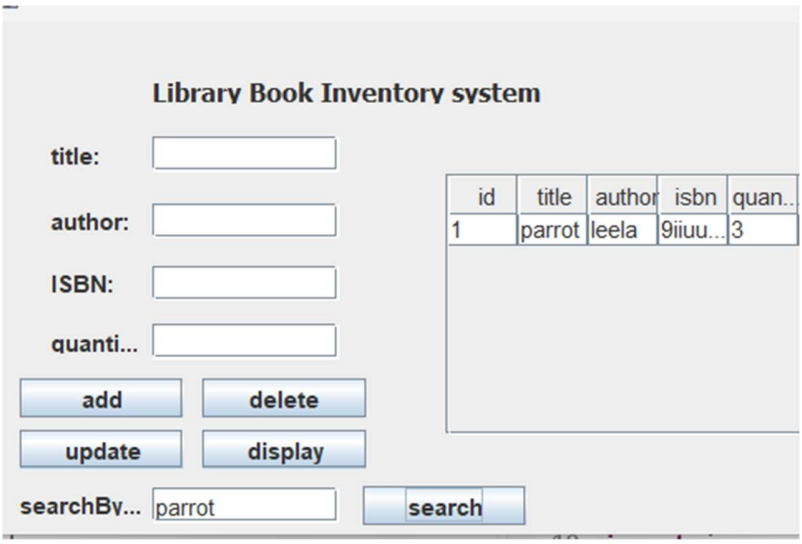
                }
            }

        });


    }


}
```





Assignment 4: Employee Management System

- **Description**: Build an employee management system with the following features:
  - Add a new employee to the database (employee ID, name, department, salary).
  - Display all employees in a JTable.

- Update employee details (e.g., salary increase, department change).
- Delete an employee record.
- Calculate the total salary of all employees and display it in a label.

```
import java.awt.EventQueue;

import javax.swing.JFrame;
import javax.swing.JPanel;
import javax.swing.border.EmptyBorder;
import javax.swing.JLabel;
import javax.swing.JOptionPane;

import java.awt.Font;
import javax.swing.JTextField;
import javax.swing.JButton;
import javax.swing.JTextArea;
import javax.swing.JScrollPane;
import javax.swing.JTable;
import javax.swing.table.DefaultTableModel;

import net.proteanit.sql.DbUtils;
```

```
import java.awt.event.ActionListener;
import java.sql.Connection;
import java.sql.DriverManager;
import java.sql.PreparedStatement;
import java.sql.ResultSet;
import java.awt.event.ActionEvent;

public class Emp extends JFrame {

	private static final long serialVersionUID = 1L;
	private JPanel contentPane;
	private JTextField t1;
	private JTextField t2;
	private JTextField t3;
	private JTextField t4;
	private JTable table;
	private JTextField saltxt;

	/**
	 * Launch the application.
	 */
	public static void main(String[] args) {
		EventQueue.invokeLater(new Runnable() {
```

```
                public void run() {
                        try {
                                Emp frame = new Emp();
                                frame.setVisible(true);
                        } catch (Exception e) {
                                e.printStackTrace();
                        }
                }
        });
}

/**
 * Create the frame.
 */
public Emp() {
        setDefaultCloseOperation(JFrame.EXIT_ON_CLOSE);
        setBounds(100, 100, 450, 300);
        contentPane = new JPanel();
        contentPane.setBorder(new EmptyBorder(5, 5, 5, 5));
        setContentPane(contentPane);
        contentPane.setLayout(null);

        JLabel lblNewLabel = new JLabel("Employee Records:");
```

```
lblNewLabel.setFont(new Font("Tahoma", Font.BOLD | Font.ITALIC, 13));
lblNewLabel.setBounds(89, 10, 124, 12);
contentPane.add(lblNewLabel);

JLabel lblNewLabel_1 = new JLabel("Id:");
lblNewLabel_1.setBounds(61, 48, 44, 12);
contentPane.add(lblNewLabel_1);

JLabel lblNewLabel_2 = new JLabel("Name:");
lblNewLabel_2.setBounds(61, 78, 44, 12);
contentPane.add(lblNewLabel_2);

JLabel lblNewLabel_3 = new JLabel("dept:");
lblNewLabel_3.setBounds(61, 110, 44, 12);
contentPane.add(lblNewLabel_3);

JLabel lblNewLabel_4 = new JLabel("sal:");
lblNewLabel_4.setBounds(61, 143, 44, 12);
contentPane.add(lblNewLabel_4);

t1 = new JTextField();
t1.setBounds(91, 45, 96, 18);
contentPane.add(t1);
```

```
t1.setColumns(10);

t2 = new JTextField();
t2.setBounds(89, 75, 96, 18);
contentPane.add(t2);
t2.setColumns(10);

t3 = new JTextField();
t3.setBounds(91, 107, 96, 18);
contentPane.add(t3);
t3.setColumns(10);

t4 = new JTextField();
t4.setBounds(89, 140, 96, 18);
contentPane.add(t4);
t4.setColumns(10);

JButton btnNewButton = new JButton("add");
btnNewButton.addActionListener(new ActionListener() {
        public void actionPerformed(ActionEvent e) {
                try {
                        Connection con=DriverManager.getConnection(
                                        "jdbc:mysql://localhost:3306/emp_db",
```

```
                                        "root",
                                        "Root01");
                                String query="insert into
emp(id,name,dept,sal)values(?,?,?,?)";
                                PreparedStatement pst=con.prepareStatement(query);
                                pst.setString(1,t1.getText());
                                pst.setString(2,t2.getText());
                                pst.setString(3,t3.getText());
                                pst.setInt(4,Integer.parseInt(t4.getText()));
                                pst.executeUpdate();
                                JOptionPane.showConfirmDialog(null, "emp added!");


                }catch(Exception e1) {
                        e1.printStackTrace();
                }

                }
        });

        btnNewButton.setBounds(10, 165, 84, 20);
        contentPane.add(btnNewButton);
```

```
JButton btnNewButton_1 = new JButton("display");
btnNewButton_1.addActionListener(new ActionListener() {
	public void actionPerformed(ActionEvent e) {
		try {
			Connection con=DriverManager.getConnection(
					"jdbc:mysql://localhost:3306/emp_db",
					"root",
					"Root01");
			String sql="select * from emp";
			PreparedStatement pst=con.prepareStatement(sql);
			ResultSet rs=pst.executeQuery();
			table.setModel(DbUtils.resultSetToTableModel(rs));

			}catch(Exception e1) {
				e1.printStackTrace();

			}
	}
});
btnNewButton_1.setBounds(10, 202, 84, 20);
contentPane.add(btnNewButton_1);

JButton btnNewButton_2 = new JButton("update");
```

```
btnNewButton_2.addActionListener(new ActionListener() {
        public void actionPerformed(ActionEvent e) {
                try {
            int row = table.getSelectedRow();


            if (row == -1) {
               JOptionPane.showMessageDialog(null, "Please select a row first!");
               return;
            }


            DefaultTableModel model = (DefaultTableModel) table.getModel();


            int iid = Integer.parseInt(model.getValueAt(row, 0).toString());


            Connection con = DriverManager.getConnection(
                  "jdbc:mysql://localhost:3306/emp_db",
                  "root",
                  "Root01");
```

```
            String sql = "UPDATE emp SET id=?,name=?, dept=?, sal=? WHERE id=?";
            PreparedStatement pst = con.prepareStatement(sql);

            pst.setString(1, t1.getText());
            pst.setString(2, t2.getText());
            pst.setString(3, t3.getText());
            pst.setInt(4, Integer.parseInt(t4.getText()));
            pst.setInt(5, iid);

            pst.executeUpdate();

            JOptionPane.showMessageDialog(null, "emp Updated Successfully!");

        } catch (Exception e1) {
            e1.printStackTrace();
        }
            }
});
btnNewButton_2.setBounds(103, 165, 84, 20);
contentPane.add(btnNewButton_2);

JButton btnNewButton_3 = new JButton("delete");
```

```
btnNewButton_3.addActionListener(new ActionListener() {
        public void actionPerformed(ActionEvent e) {
                try {
            int row = table.getSelectedRow();


            if (row == -1) {
               JOptionPane.showMessageDialog(null, "Please select a row first!");
               return;
            }


            DefaultTableModel model = (DefaultTableModel) table.getModel();


            int iid = Integer.parseInt(model.getValueAt(row, 0).toString());


            Connection con = DriverManager.getConnection(
                  "jdbc:mysql://localhost:3306/emp_db",
                  "root",
                  "Root01");
```

```
            String sql = "delete from emp WHERE id=?";
            PreparedStatement pst = con.prepareStatement(sql);
            pst.setInt(1, iid);

            pst.executeUpdate();

            JOptionPane.showMessageDialog(null, "emp deleted Successfully!");

        } catch (Exception e1) {
            e1.printStackTrace();
        }

            }
});
btnNewButton_3.setBounds(103, 202, 84, 20);
contentPane.add(btnNewButton_3);



table = new JTable();
table.setModel(new DefaultTableModel(
            new Object[][] {
```

```
		},
		new String[] {
			"id", "name", "dept", "sal"
		}
));
table.setBounds(236, 47, 190, 167);

JScrollPane scrollPane = new JScrollPane(table);
scrollPane.setBounds(242, 46, 184, 167);
contentPane.add(scrollPane);

JLabel lblNewLabel_5 = new JLabel("Total_sal");
lblNewLabel_5.setBounds(36, 241, 58, 12);
contentPane.add(lblNewLabel_5);

JButton btnNewButton_4 = new JButton("total");
btnNewButton_4.addActionListener(new ActionListener() {
	public void actionPerformed(ActionEvent e) {
		try {
	Connection con = DriverManager.getConnection(
		"jdbc:mysql://localhost:3306/emp_db", "root", "Root01");

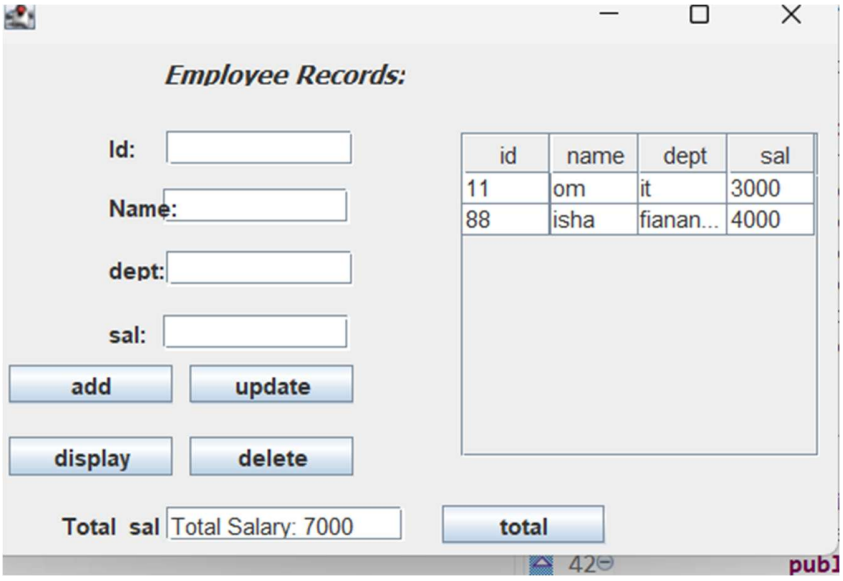
	String sql = "SELECT SUM(sal) FROM emp";
```

```
                PreparedStatement pst = con.prepareStatement(sql);
                ResultSet rs = pst.executeQuery();

                if (rs.next()) {
                    int total = rs.getInt(1);
                    saltxt.setText("Total Salary: " + total);
                }

            } catch (Exception ex) {
                ex.printStackTrace();
            }
        }
    });
    btnNewButton_4.setBounds(232, 237, 84, 20);
    contentPane.add(btnNewButton_4);

    saltxt = new JTextField();
    saltxt.setBounds(91, 238, 122, 18);
    contentPane.add(saltxt);
    saltxt.setColumns(10);
  }
}
```



Assignment 5 : Invoice Generation System

- **Description**: Create an invoice generation system for a shop where customers can place orders. The system should:
    - Allow the user to add multiple items (product name, price, quantity).
    - Calculate the total cost of the items (price * quantity).
    - Generate and save the invoice to the database.
    - Display a list of previous invoices in a JTable with options to view individual invoices.

```
import java.awt.EventQueue;


import javax.swing.JFrame;

import javax.swing.JPanel;

import javax.swing.border.EmptyBorder;
```

```
import javax.swing.JLabel;
import javax.swing.JOptionPane;

import java.awt.Font;
import javax.swing.JTextField;
import javax.swing.JButton;
import javax.swing.JScrollPane;
import javax.swing.JTable;
import javax.swing.table.DefaultTableModel;

import net.proteanit.sql.DbUtils;

import java.sql.*;

public class Invo extends JFrame {

	private JPanel contentPane;
	private JTextField t1, t2, t3;
	private JTable table;
	private JTextField totalTxt;

	public static void main(String[] args) {
		EventQueue.invokeLater(() -> {
```

```
                try {
                    Invo frame = new Invo();
                    frame.setVisible(true);
                } catch (Exception e) {
                    e.printStackTrace();
                }
            });
    }

    public Invo() {
        setDefaultCloseOperation(JFrame.EXIT_ON_CLOSE);
        setBounds(100, 100, 500, 350);
        contentPane = new JPanel();
        contentPane.setBorder(new EmptyBorder(5, 5, 5, 5));
        setContentPane(contentPane);
        contentPane.setLayout(null);

        JLabel lblTitle = new JLabel("Invoice System");
        lblTitle.setFont(new Font("Tahoma", Font.BOLD, 14));
        lblTitle.setBounds(160, 10, 150, 20);
        contentPane.add(lblTitle);

        JLabel l1 = new JLabel("Product:");
```

```
l1.setBounds(20, 50, 80, 20);
contentPane.add(l1);

JLabel l2 = new JLabel("Price:");
l2.setBounds(20, 80, 80, 20);
contentPane.add(l2);

JLabel l3 = new JLabel("Qty:");
l3.setBounds(20, 110, 80, 20);
contentPane.add(l3);

t1 = new JTextField();
t1.setBounds(90, 50, 100, 20);
contentPane.add(t1);

t2 = new JTextField();
t2.setBounds(90, 80, 100, 20);
contentPane.add(t2);

t3 = new JTextField();
t3.setBounds(90, 110, 100, 20);
contentPane.add(t3);
```

```
// JTable
table = new JTable();
table.setModel(new DefaultTableModel(
        new Object[][] {},
        new String[] {"Product", "Price", "Qty", "Total"}
));

JScrollPane scrollPane = new JScrollPane(table);
scrollPane.setBounds(220, 50, 240, 150);
contentPane.add(scrollPane);

// ADD ITEM
JButton btnAdd = new JButton("Add Item");
btnAdd.setBounds(20, 150, 120, 25);
contentPane.add(btnAdd);

btnAdd.addActionListener(e -> {
        try {
                String name = t1.getText();
                int price = Integer.parseInt(t2.getText());
                int qty = Integer.parseInt(t3.getText());

                int total = price * qty;
```

```
                DefaultTableModel model = (DefaultTableModel)
table.getModel();
                model.addRow(new Object[]{name, price, qty,
total});

            } catch (Exception ex) {
                JOptionPane.showMessageDialog(null, "Enter valid
data!");
            }
        });

        // TOTAL
        JButton btnTotal = new JButton("Total");
        btnTotal.setBounds(20, 190, 120, 25);
        contentPane.add(btnTotal);

        totalTxt = new JTextField();
        totalTxt.setBounds(150, 190, 100, 25);
        contentPane.add(totalTxt);

        btnTotal.addActionListener(e -> {
            int sum = 0;
            for (int i = 0; i < table.getRowCount(); i++) {
```

```
                    sum += Integer.parseInt(table.getValueAt(i,
3).toString());
                }
                totalTxt.setText(String.valueOf(sum));
            });

            // GENERATE INVOICE
            JButton btnSave = new JButton("Save Invoice");
            btnSave.setBounds(20, 230, 150, 25);
            contentPane.add(btnSave);

            btnSave.addActionListener(e -> {
                try {
                    Connection con = DriverManager.getConnection(
                            "jdbc:mysql://localhost:3306/invo",
"root", "Root01");

                    // calculate total
                    int grandTotal = 0;
                    for (int i = 0; i < table.getRowCount(); i++) {
                        grandTotal +=
Integer.parseInt(table.getValueAt(i, 3).toString());
                    }
```

```
                    // insert invoice
                    String sql = "INSERT INTO invoices(total_amount)
VALUES(?)";
                    PreparedStatement pst =
con.prepareStatement(sql, Statement.RETURN_GENERATED_KEYS);
                    pst.setInt(1, grandTotal);
                    pst.executeUpdate();

                    ResultSet rs = pst.getGeneratedKeys();
                    rs.next();
                    int invoiceId = rs.getInt(1);

                    // insert items
                    for (int i = 0; i < table.getRowCount(); i++) {
                        String name = table.getValueAt(i,
0).toString();
                        int price =
Integer.parseInt(table.getValueAt(i, 1).toString());
                        int qty = Integer.parseInt(table.getValueAt(i,
2).toString());
                        int total =
Integer.parseInt(table.getValueAt(i, 3).toString());
```

```
                    String itemSql = "INSERT INTO
invoice_items(invoice_id, product_name, price, quantity, total) VALUES (?, ?, ?, ?,
?)";

                    PreparedStatement pst2 =
con.prepareStatement(itemSql);


                    pst2.setInt(1, invoiceId);

                    pst2.setString(2, name);

                    pst2.setInt(3, price);

                    pst2.setInt(4, qty);

                    pst2.setInt(5, total);


                    pst2.executeUpdate();

                }


                    JOptionPane.showMessageDialog(null, "Invoice
Saved!");


                    // clear table

                    ((DefaultTableModel)
table.getModel()).setRowCount(0);


                } catch (Exception ex) {

                    ex.printStackTrace();
```

```
            }
        });

        // VIEW INVOICES
        JButton btnView = new JButton("View Invoices");
        btnView.setBounds(180, 230, 150, 25);
        contentPane.add(btnView);

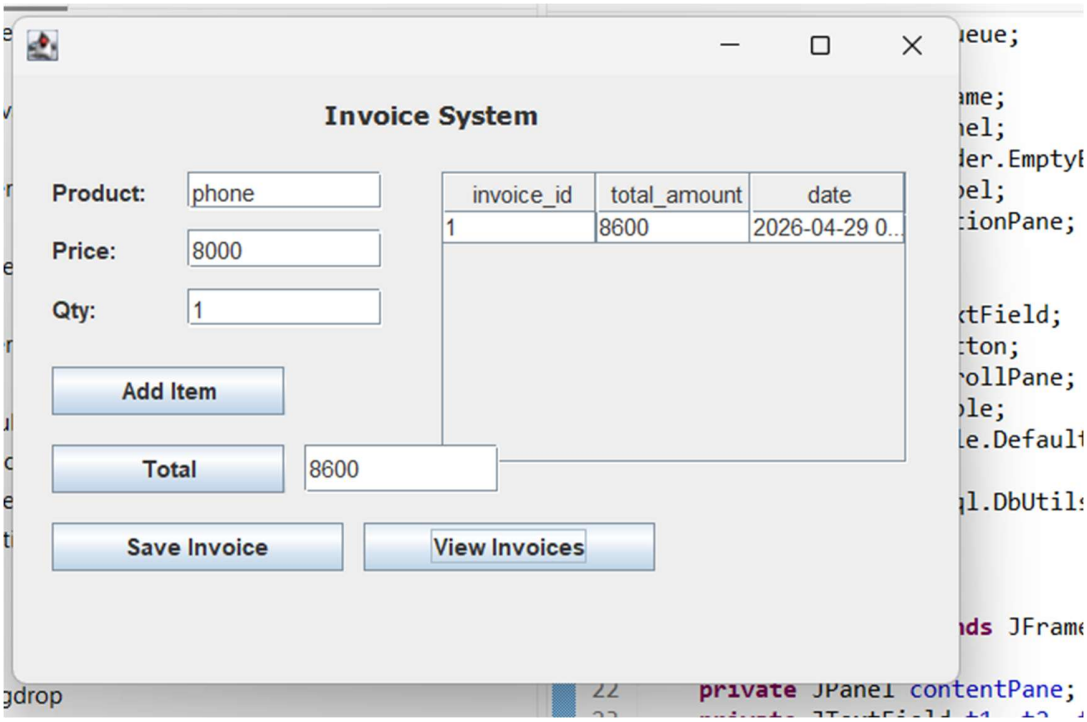
        btnView.addActionListener(e -> {
            try {
                Connection con = DriverManager.getConnection(
                        "jdbc:mysql://localhost:3306/invo",
"root", "Root01");

                String sql = "SELECT * FROM invoices";
                PreparedStatement pst =
con.prepareStatement(sql);
                ResultSet rs = pst.executeQuery();

                table.setModel(DbUtils.resultSetToTableModel(rs));

            } catch (Exception ex) {
                ex.printStackTrace();
```

```
                }
            });
        }
    }
```


Invoice System
Product:
phone
Price:
8000
Qty:
1
Add Item
Total
8600
Save Invoice
View Invoices
invoice_id
total_amount
date
1
8600
2026-04-29 0...